

EE 210L: Digital Circuits and Logic Design Lab

Lab 4: Karnaugh Maps

Name: _____

Date: _____

Partner(s): _____

Section: _____

Record logic levels only, 0 or 1. The map labels are already in Gray code order, 00 · 01 · 11 · 10. Mark every cell, then circle every group you use.

Part 4: Maps from the Pre-Lab

4.1 Majority voter M(A,B,C)

A\BC	00	01	11	10
0				
1				

Minimal SOP from the map M = _____

Does it agree with the expression you derived with the laws in Lab 3? Which method took fewer steps?

4.2 Function W(a,b,c,d)

ab\cd	00	01	11	10
00				
01				
11				
10				

Don't-cares marked X and used.

ab\cd	00	01	11	10
00				
01				
11				
10				

Every X forced to 0.

Don't-cares used W = _____

Every X forced to 0 W = _____

4.3 Prime detector P(a,b,c,d)

ab\cd	00	01	11	10
00				
01				
11				
10				

Eight cells cannot occur. Mark them X.

The eight codes that cannot occur: _____

All three minimal forms of P: _____

Part 5.1: Power Supply Setup

Measured power supply voltage: _____ V

Current limit setting: _____ mA

Part 5.2: Function W, What the Don't-Cares Bought You

Expression built W = _____

#	a	b	c	d	Predicted W	Measured W
1	0	0	0	0		
2	0	0	1	0		
3	0	0	1	1		
4	0	1	0	0		
5	0	1	0	1		
6	0	1	1	0		
7	0	1	1	1		
8	1	0	0	0		
9	1	0	1	0		
10	1	0	1	1		
11	1	1	0	1		
12	1	1	1	0		

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The four rows below cannot occur in the system this function came from. Record what the circuit does anyway.

a	b	c	d	Measured W
0	0	0	1	
1	0	0	1	
1	1	0	0	
1	1	1	1	

Why were you free to ignore these four rows when you chose your groups?

Part 5.3: Function P, the Prime Detector

Expression built P = _____

#	Digit	a	b	c	d	Minterm	Predicted P	Measured P
1	0							
2	1							
3	2							
4	3							
5	4							
6	5							
7	6							
8	7							

Does the measured column match the specification: 1 for digits 1, 2, 3, 5, 7 and 0 for digits 0, 4, 6?

Part 5.4: Minimum Is Not Unique

Second form built P = _____

Digit	0	1	2	3	4	5	6	7
P from §5.3								
P from §5.4								

The two circuits agree on all eight valid digits. Do they also agree on the eight codes that cannot occur? Explain the result.

Part 5.5: Cost Comparison

Count 2-input gates. An n -input gate costs $n - 1$ of them. Count inverters separately.

Function	Version	Terms	AND	OR	NOT	Total gates	Packages
W	Don't-cares used						
W	X forced to 0						
P	Don't-cares used						
P	X forced to 0						

Would the all-zeros version of P fit the chips you brought? _____

Discussion and Conclusions

Summarize what you built and what it confirmed. Address at least: what the map does that the laws of Lab 2 cannot do, why a don't-care costs nothing to use, and what it means for two different expressions to be equally minimal.